

CHAPTER 1

The Ultrarunning Revolution

As this book finds its way into your eager hands, I will be well into my third decade of being a professional coach. During that time, I've worked with thousands of individual (and very different) athletes. I've helped guide marathon runners, cyclists, triathletes, ultrarunners, mixed martial arts fighters, motorsports athletes (NASCAR, motocross, and Formula One), various winter sports athletes, and a host of other athletes looking to develop their endurance. Some of those sports were familiar to me at the outset. As a lifelong runner, I had a background in the traditional distance events (5K, 10K, and marathon) and could readily adapt to other sports like cycling and triathlon. Running, cycling, and triathlon also had rich histories of peer-reviewed physiological research, as well as books like Daniels' Running Formula to draw upon. In working with any moderately fit marathoner, I could use a combination of my previous running experience, scientific literature, and other coaches' guidance to comprehensively inform training. Thankfully, as a young coach I wound up in an environment that forced me outside sports I was familiar with, and that has made all the difference.

You learn the most when you are out of your comfort zone, and working with MMA fighters and NASCAR drivers was way out of my comfort zone. I didn't know anything about those sports, which makes it pretty hard to improve athlete performance based on sheer experience. These sports also had relatively little scientific literature to draw upon and few points of reference to inform training. As a result, I was forced to learn about the sports from the ground up. In order to figure out how endurance training could be applied to sports not traditionally mentioned alongside marathon running, cycling, or triathlon, I developed a method of first meticulously learning about the sport before diving into the x's and o's of coaching. I would peel apart the physiological and psychological demands of the sport using a combination of old-school observation and best practices borrowed from similar sports.

When our coaching team started working with NASCAR drivers, we'd never been inside of a race car, never attended a race, and didn't even know the basic rule structure. So, our first order of business was to attend a routine practice alongside the engineers, pit crew, and the rest of the support system required to operate a NASCAR team. We rigged the drivers up with heart rate monitors and observed their breathing patterns over race radio to determine the cardiorespiratory stress during a race. It turns out that their heart and lungs are working as hard as any marathoner, so we designed training strategies for these athletes that leveraged proven training methods from other endurance disciplines. We had them train like marathoners and triathletes, complete with intervals, threshold runs, and long runs. We measured the temperatures in the car and inside the drivers' specially designed suits and quickly determined that the conditions are similar to a summer day in Death Valley (which ironically would be another environment I became quite familiar with). As a result, we also had them perform heat stress protocols in the sauna, not too dissimilar to any athlete preparing for a hot-weather event like the Western States 100 or Badwater 135. And you thought all there was to NASCAR racing was to sit down and turn left.

The results provided proof that endurance training techniques can improve performance in sports outside the traditional endurance sports. Reducing the physical strain on the drivers though improved cardiovascular fitness and heat tolerance improved their driving. The drivers were able to better control their cars and communicate with their team because fewer of their physiological resources were dedicated to fighting the demands of the event. All of these interventions were developed through observation and borrowing protocols from other sports, a tactic that would serve me well as I entered into the realm of ultramarathon coaching.

As ironic as it might sound now, ultrarunners were actually very late to the table when it came to adopting and accepting coaching. And it was not from a lack of trying. In the early 2000s I spent frustrating amounts of time trying to convince elite ultrarunners that coaching would be a good thing for them. Ultrarunning is a complicated affair, so in my mind it seemed clear that some professional guidance would do the sport's elite athletes some good. Alas, my pitches fell on deaf ears in those early years, and I got laughed out of more than several meetings.

Finally, in 2004 some of these athletes started to turn the corner and seek out professional coaching. Being that this was a new sport group to work with, I did what I always did with a new sport: I dug through copious amounts of research and looked for guidance. Except in this case, there was little to be found. The same year I started working with ultrarunners, there were exactly six research papers on the subject. No books on ultramarathon training existed. Any guidance I could find had to be mined from blogs, anecdotes, and the infamous ultra listserv (a message board for ultrarunners that the old-timers will remember fondly). So, absent any scientific guidance, just like with the NASCAR drivers, I went and observed. I went to races, paced athletes, spoke with those who were successful, and started running ultramarathons myself, all in an effort to understand the sport better. I also borrowed best practices from cycling, triathlon, and even cross-country skiing, because in many cases the demands from those sports overlap with ultrarunning.

Developing training strategies through observation and educated guessing works to an extent, particularly when it is all you have. And I have to say, the athletes I worked with in those early years did far better than anticipated. They thrived, won races around the globe, and achieved success that far outpaced their peers. People started to notice, so much so that a cottage industry for ultrarunning coaching sprouted and blossomed. As the sport grew and elite performances improved, the pace of research into what constitutes ultramarathon performance increased. All of this was good for athletes. Better information led to evolutions in coaching practices across the entire sport and more information for athletes to use on their own . . . but it was scattered. I wanted a way to consolidate it all. I wanted something that would present the training methodologies I had observed and curated side by side with emerging research. I wanted something that could synthesize best practices, the art of coaching, and emerging science to provide a comprehensive guide for ultramarathon athletes and their coaches. And for the first fifteen years I coached ultrarunners, this amalgamation failed to emerge.

The first edition of this book was written, in large part, to create the book I wanted as a coach and athlete. I sought to create a reference that would inspire, inform, and guide ultrarunners and coaches along their journeys. I wanted to create a book that I personally would have turned to when I first started working with ultrarunners, much like I used Daniels' Running Formula and The Lore of Running early in my coaching career. By any measuring stick, that effort was a success. Ultrarunners all along the spectrum of novices to elites finally had a book that blended science and practice. They had a manual to guide them, regardless of whether they were training for a 50K in the desert, 200 miles in the mountains, or any distance and terrain in between. Tens of thousands of copies of the first edition of this book have been sold. It's been translated into multiple languages, and I've traveled around the world to give lectures on its contents. I've even had to recruit, screen, and train an entire ultrarunning coaching department (which at the time this book is published sits twelve coaches strong) to satiate the demand for athletes wanting to be coached.

Despite that success, the first edition of this book had some detractors. The training methods came under scrutiny, as they rightfully should, and at times by influential and well-known coaches. In 2016, just before the first edition of this book went to press, I was profiled by Trail Runner Magazine for a feature in their annual "Dirt" edition. The feature contained the normal Q&A-style interview, complete with a tour of my office and physiology lab, as well as interviews with many of the athletes I have worked with. Unbeknownst to me at the time, the author also gathered the opinions of trail running coaches David Roche, Ian Torrence, and Ian Sharman. Mr. Roche described some of the training strategies I use as chasing "marginal gains," which is a contentious strategy that aims to compound many small incremental improvements to potentially create a larger meaningful benefit. Mr. Torrence and Mr. Sharman were critical of some of the strategic underpinnings I use for ultrarunners that are derived from different sports such as cycling.

Since the first edition of this book was published, some of those criticisms have echoed. I've been told "you don't believe in strength training," "ultrarunners are not cross-country skiers," and the like. The fact of the matter is that most of these criticisms are ill-informed, shortsighted, poorly thought out, and lacking context. For example, anyone who has followed the hundreds of articles, interviews, and other pieces of content I have posted over my career knows that I preach a consistent theme of focusing on what matters most for improvement before chasing any "marginal gains." Additionally, there are certain, limited circumstances where many marginal gains can mean the difference between an athlete winning and losing, or making a cutoff or not. Ultramarathon training is highly nuanced and rewards coaches and athletes who are open to new information and willing to accept that there may be multiple paths to a successful outcome.

The contents of this book exemplify this way of thinking. As you will come to find out, you can weigh the benefits you get from a sauna, strength training, or any other "marginal" intervention against one another and make choices on which gains to chase and which to forgo. Furthermore, while ultrarunning is a unique sport, it is

not so dissimilar from cycling, cross-country skiing, triathlon, marathon running, or any other endurance sport that we can't borrow and adapt research and best practices to better train ultramarathon runners. The fact of the matter is, there is far more information on training athletes in those traditional endurance sports than training ultrarunners, and there likely always will be. The key is finding the nuance in what can be applied across different endurance disciplines, what needs to be adapted from one sport to another, and what doesn't apply at all.

Our understanding of ultramarathon running continues to evolve, and this book had to evolve as well. We know far more now about ultramarathon performance than we did at the time of the original publication. We've learned more about concepts like how neuromuscular fatigue likely plays an important role in ultramarathon performances, low-carbohydrate diets impact running economy, and hyponatremia (low blood sodium) is more often caused by overdrinking fluids that do not contain sodium than not consuming enough sodium. The six scientific papers published in 2004 have now expanded to more than one hundred per year, providing a growing trove of information coaches and athletes have the opportunity to digest and synthesize into their training.

With that as a backdrop, here's what's new in the second edition of *Training Essentials for Ultrarunning*: almost everything. In fact, more than 75 percent of the content has been revised, expanded, or written anew. Among other additions, there is a section devoted to low-carbohydrate diets, an entire chapter on the benefits (or lack thereof) of strength training, a chapter on mental skills for ultrarunning, and an in-depth section devoted specifically to considerations for female athletes. And to make sure I, as a male coach, wasn't unconsciously biasing the recommendations for female athletes, as has often happened in various areas of research, I brought in my coaching colleague Corrine Malcolm to specifically pen that section. This book also went through greater scientific scrutiny from Nick Tiller, PhD, and Stephanie Howe, PhD, both ultrarunners who are well respected in their fields. This expands upon the scientific guidance provided by preeminent biomechanist Rodger Kram, PhD, and former Western States medical director Martin Hoffman, MD, in the first edition of this book. There are nearly 400 references to scientific literature, all of which can be found in the book's references section.

Despite the additions, you still will not find a stock training plan in this book. If you are expecting to see a page you can tear out, put on your refrigerator, and blindly follow for the next twenty weeks, you are going to be sorely disappointed. So, don't say I didn't warn you, and please ask for your money back now before you leave a one-star Amazon review. Better yet, read the book, and at the end, if you hate it, email me and I will gladly refund you, no questions asked.

This book will walk you through the steps of solving your personal puzzle of ultramarathon performance. It will take you on a journey, step by step, on how ultramarathons work, what you can do to better prepare for them, and how you can thrive during the process. At the end of that journey, I want you to come away well informed and confident about whatever ultramarathon adventure you have chosen. I want you to understand what it takes to make yourself a better ultrarunner. I want you to have enough information and wisdom to create and adjust your own training, nutrition, and ultramarathon game plan, and to be confident with the process. I want you to dog-ear the pages in this book that have particular meaning to you and scribble notes in the margins. Above all else, I want you to be inspired by the training processes you uncover and the realization that you are ready to crush your most audacious goals. After all, ultrarunning is a ridiculously difficult sport, one that is rife with pitfalls and hazards that confound even the most seasoned athletes. Preparation is your greatest ally and most powerful resource, and the information in this book will help you be as prepared as possible when you step up to the start line.

Finally, I would be remiss not to mention that this book does not have any sponsors, endorsers, or other sources of financial or business entanglement. In 2018, I made the conscious decision to alleviate myself from any such conflicts of interest, real or perceived, and I have to say that the move was quite liberating. In today's world of paid partnerships, Instagram influencers, and implied endorsements, it is nearly impossible for the end consumer to ascertain fact from fiction. While there are many athletes and influencers that authentically use and endorse the products they are peddling, there are just as many that wear a different brand of shoes on their feet, have a different drink in their bottle, or are just taking the endorsement money and running. So, I decided to avoid these conflicts altogether and turned down every endorsement opportunity that has come my way since (much to the dismay of my bank account). As a consequence, if I recommend something it is because I believe in the product or method based on its merits, not because someone is paying me to do so. This freedom also allows me to opine on products and practices that are, quite frankly, junk and pseudoscientific garbage. And I certainly don't mind calling them out as such.

To take it a step further, I acquired the rights to self-publish this book in order to make it as long, detailed, and accurate as possible. (Plus I wanted to say whatever I pleased, which is problematic with even the most liberal of

publishers.) Make no mistake, these actions cost me a copious amount of time and a fair amount of money. I hired and paid editors, contributors, printers, and the like out of my own pocket in order to get this edition of the book off of the ground. But the trade-off is worth it, and I wouldn't change a thing. In my profession, I am ultimately a purveyor of advice. Athletes trust in the words I write and say. Based on that trust, they take action from those words. I honor that trust as a sacred bond between me and the athletes reading these pages. So, rest assured the content that follows contains the truth, the whole truth, and nothing but the truth in a pure, raw, unentangled, and unadulterated fashion all the way down to the last sentence.

So, if you are ready, willing, and able, let's get right into it!



